Cuantificación de Incertidumbre en Clasificación Histopatológica

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Abstract

La adopción clínica de sistemas de aprendizaje profundo en anatomía patológica se ve limitada no tanto por la precisión de los modelos, sino por la incapacidad de estos de señalar cuándo no están seguros de sus predicciones. En este trabajo se comparan empíricamente tres modelos (dos de ellos con cuantificación de incertidumbre) que parten del mismo clasificador de referencia (ResNet18 preentrenada) aplicado al dataset *Histopathologic Cancer Detection*: (i) una red determinista entrenada como estimador máximo a posteriori (MAP); (ii) una aproximación de Laplace *post-hoc* sobre la última capa, con Hessiano factorizado por Kronecker (KFAC) y precisión del prior optimizada por verosimilitud marginal; y (iii) Monte Carlo Dropout con $T{=}50$ pasadas estocásticas en inferencia. Los tres modelos se evalúan en términos de discriminación (AUC-ROC, F1), calibración probabilística (Expected Calibration Error, Brier Score, log-verosimilitud negativa) y utilidad clínica mediante un análisis de triaje basado en incertidumbre epistémica.

1 Introducción

1.1 Motivación clínica y valor de negocio

La detección automática de células tumorales en parches de tejido histopatológico es uno de los casos de uso más consolidados del aprendizaje profundo en medicina. Los modelos convolucionales actuales alcanzan sistemáticamente un AUC superior a $0{,}95$ sobre benchmarks públicos, lo que ha motivado su integración progresiva en flujos de cribado asistido por ordenador. Sin embargo, este rendimiento oculta un problema estructural: un clasificador determinista emite una probabilidad puntual $\hat{p}(y{=}1 \mid \mathbf{x})$ sin ninguna cuantificación de la fiabilidad de esa estimación. Un caso morfológicamente ambiguo y un caso prototípico pueden recibir la misma probabilidad sin que el sistema distinga entre ellos.

Esta limitación es grave en dominios donde el coste de un falso negativo es desproporcionado y crítico, como puede ser la oncología. En este contexto, el enfoque bayesiano aporta una solución práctica: en lugar de aprender un único conjunto de pesos $\boldsymbol{\omega}^*$, el objetivo es inferir una distribución posterior $p(\boldsymbol{\omega} \mid \mathcal{D})$ sobre la que se marginaliza al predecir. La varianza de dicha distribución predictiva posibilita la toma de decisiones diferenciadas: cuando la incertidumbre supera un umbral preconfigurado τ_u, la muestra se deriva a revisión por un patólogo.

Esta regla operativa es la traducción directa del valor de negocio de estos modelos: no se trata solo de mejorar la precisión global, sino de identificar casos inciertos para que un experto (humano) los revise, reduciendo así el riesgo de falsos negativos sin sobrecargar al patólogo con casos típicos. En un despliegue real, las métricas online asociadas serían la tasa de derivación a revisión humana, la frecuencia con la que el patólogo rechaza la recomendación del modelo teniendo visibilidad de la incertidumbre (*user trust*), y la *reducción de falsos negativos* de la combinación modelo-patólogo frente a un patólogo sin apoyo algorítmico.

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1.2 Objetivos y contribuciones

Este trabajo persigue tres objetivos concretos. Primero, comparar cuantitativamente el rendimiento predictivo y la calibración de dos métodos de inferencia aproximada ampliamente utilizados en la literatura: la aproximación de Laplace (**?**) y Monte Carlo (MC) Dropout (**?**). En ambos casos, se parte de un mismo clasificador fijo (ResNet18 preentrenada) para aislar el efecto del método probabilístico. Segundo, analizar y aplicar la descomposición de la incertidumbre predictiva en sus componentes aleatórica y epistémica para definir una política de triaje. Tercero, estudiar los recursos computacionales de cada enfoque (tiempo de entrenamiento, latencia de inferencia, memoria) para contemplar un posible despliegue. El código, los pesos entrenados y una demostración interactiva en Streamlit están disponibles en este repositorio, reproducible con semilla aleatoria fija.

2 Formulación del problema

2.1 Modelo probabilístico

Disponemos de un conjunto de datos (dataset *Histopathologic Cancer Detection*) $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ con imágenes $\mathbf{x}_i \in \mathbb{R}^{3\times 96\times 96}$ y etiquetas binarias $y_i \in \{0, 1\}$ que indican la presencia o ausencia de tejido tumoral la región central (32x32) de la imagen. Modelizamos la verosimilitud como categórica sobre dos clases,

$$p(y \mid \mathbf{x}, \boldsymbol{\omega}) \;=\; \mathrm{Cat}\big(y \,\big|\, \mathrm{softmax}(f_{\boldsymbol{\omega}}(\mathbf{x}))\big), \tag{1}$$

donde $f_{\boldsymbol{\omega}} : \mathbb{R}^{3\times 96\times 96} \to \mathbb{R}^2$ es una red neuronal con pesos $\boldsymbol{\omega} \in \mathbb{R}^P$. Aunque el problema es binario, tomamos la parametrización *softmax* de dos dimensiones por cuestiones de compatibilidad con la librería `laplace-torch` (que asume salida categórica) y con el mecanismo de descomposición de la incertidumbre explicado en la SECCION4!.

Asumimos un prior gaussiano isotrópico sobre los pesos,

$$p(\boldsymbol{\omega}) \;=\; \mathcal{N}\big(\boldsymbol{\omega} \,\big|\, \mathbf{0}, \lambda^{-1}\mathbf{I}\big), \tag{2}$$

con precisión $\lambda > 0$. Para obtener la distribución predictiva bayesiana para una nueva observación $\mathbf{x}^\star$ necesitamos marginalizar sobre el posterior,

$$p(y^\star \mid \mathbf{x}^\star, \mathcal{D}) \;=\; \int p(y^\star \mid \mathbf{x}^\star, \boldsymbol{\omega})\, p(\boldsymbol{\omega} \mid \mathcal{D})\, \mathrm{d}\boldsymbol{\omega}. \tag{3}$$

Para ResNet18, con $P \sim 10^7$ parámetros, ni el posterior $p(\boldsymbol{\omega} \mid \mathcal{D})$ ni la integral (3) admiten forma cerrada. La metodología de este trabajo se centra en cómo aproximar esa integral manteniendo el mayor número posible de garantías bayesianas a un coste computacional aceptable.

2.2 Requisitos del sistema

Requisitos funcionales. El sistema debe (i) emitir, para cada entrada, una distribución predictiva $p(y \mid \mathbf{x})$ en lugar de una estimación puntual; (ii) descomponer la incertidumbre en sus componentes aleatórica (inherente al ruido de los datos, irreducible) y epistémica (asociada al conocimiento limitado del modelo, reducible con más datos); y (iii) producir derivar una muestra a revisión humana cuando la incertidumbre epistémica de su predicción supere el umbral preconfigurado τ_u.

Requisitos no funcionales. Establecemos como objetivos cuantitativos (i) un *Expected Calibration Error* (ECE) por debajo de 0.05; (ii) una latencia de inferencia por imagen que sea compatible con la posible integración en una interfaz de un programa médico; y (iii) un consumo de memoria que permita la ejecución en hardware de una clínica u hospital estándar. Estos requisitos condicionan, como veremos en la SECCION2! la elección del subconjunto de pesos sobre el que aplicar Laplace y la estructura adoptada para el Hessiano.

2.3 Selección de métodos

La literatura ofrece un rango amplio de métodos de inferencia aproximada para redes neuronales bayesianas (BNNs). La Tabla 1 resume los factores que más comprometen la selección, la cual debe ponderar el coste de entrenamiento, el coste de inferencia, la necesidad o no de reentrenamiento y la fidelidad de la aproximación al posterior.

Table 1: Compromisos entre métodos de inferencia aproximada en BNN. T denota el número de muestras en inferencia y M el tamaño de un *ensemble.*

Método	Coste entren.	Coste inf.	Reentren.	Cobertura del posterior
Laplace (*post-hoc*)	$1\times + 1$ *fit*	T	No	Local, unimodal
MC Dropout	$1\times$	T	Sí	Local, unimodal
Inferencia variacional (MFVI)	$\sim 2\times$	T	Sí	Unimodal
Deep Ensembles	$M\times$	M	Sí	Multimodal
MCMC (SGLD, HMC)	$\gg 1\times$	T	Sí	Asintóticamente exacta

SEGUIR DESDE AQUÍ

Seleccionamos **Laplace** y **MC Dropout** como representantes complementarios del espectro. El primero es *post-hoc* y *principled* —aprovecha la curvatura real del *loss* en el MAP— mientras que el segundo es trivial de implementar sobre cualquier arquitectura que ya incluya dropout. Ambos comparten la misma arquitectura troncal y, en el caso de Laplace, el mismo punto MAP, lo que permite atribuir las diferencias observadas casi exclusivamente al mecanismo de cuantificación de incertidumbre. Deep Ensembles y MCMC quedan fuera del alcance por coste computacional; MFVI se descarta por exigir una reimplementación completa del entrenamiento sin ventaja teórica clara sobre Laplace en términos de fidelidad local al posterior.

3 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 5 regarding figures, tables, acknowledgments, and references.

4 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

4.1 Headings: second level

Second-level headings should be in 10-point type.

4.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

5 Citations, figures, tables, references

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The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated…

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2025` package:

```
\PassOptionsToPackage{options}{natbib}
```

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```
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```

As submission is double blind, refer to your own published work in the third person. That is, use "In the previous work of Jones et al. [4]," not "In our previous work [4]." If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form "A. Anonymous" and include a copy of the anonymized paper in the supplementary material.

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Note that footnotes are properly typeset *after* punctuation marks.[2]

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All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for first word and proper nouns); figures are numbered consecutively.

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[1] Sample of the first footnote.

[2] As in this example.

Figure 1: Sample figure caption.

Table 2: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

5.4 Tables

All tables must be centered, neat, clean and legible. The table number and title always appear before the table. See Table 2.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 2.

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Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using `$$` anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

5.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

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- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

6.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

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Acknowledgments and Disclosure of Funding

Use unnumbered first level headings for the acknowledgments. All acknowledgments go at the end of the paper before the list of references. Moreover, you are required to declare funding (financial activities supporting the submitted work) and competing interests (related financial activities outside the submitted work). More information about this disclosure can be found at: `https://neurips.cc/Conferences/2025/PaperInformation/FundingDisclosure`.

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References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Technical Appendices and Supplementary Material

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